

Automatic Differentiation

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Abstract—This paper provides an explanation of Automatic Differentiation (AD) concept for university students at near-bachelor level. Includes comparison with other differentiation methods, very simple AD implementation example and more realistic usage in python library.

Index Terms—autodiff, python, computational mathematics

I. INTRODUCTION

[1], [2]

REFERENCES

- [1] Wikipedia contributors, “Automatic differentiation — Wikipedia, The Free Encyclopedia.” [Online]. Available: https://en.wikipedia.org/w/index.php?title=Automatic_differentiation&oldid=1339245165
- [2] Andrew Holmes, “What's Automatic Differentiation?.” [Online]. Available: <https://huggingface.co/blog/andmholm/what-is-automatic-differentiation>